A LOCALLY ALGEBRAIC GROUP SCHEME OVER A FIELD IS EQUIDIMENSIONAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma morphisms-lemma-dimension-fibre-after-base-change (canonical tag 02FY). cited at proof lines 1460-1462. Dimensions survive field extension, enabling rationality of the two points.
- Lemma morphisms-lemma-dimension-fibre-at-a-point (canonical tag 02FX). cited at proof lines 1466-1469. Local dimension formula.
- Lemma properties-lemma-codimension-local-ring (canonical tag 02IZ). cited at proof lines 1469-1472. Dimension as supremum of local dimensions.
- Section morphisms-section-points-finite-type (canonical tag 01T9). cited at proof lines 1473-1475. Nullstellensatz: residue fields of closed points are algebraic.

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Cited at proof lines 1460-1462 to allow extending the ground field; load-bearing for the reduction.

Source reference. Original lemma 02FX: sources/stacks-chow/tags.txt, source lines 3208–3208. Cited at proof lines 1466-1469 for the local dimension formula; load-bearing.

Source reference. Original section 01T9: sources/stacks-chow/tags.txt, source lines 2366–2366. Cited at proof lines 1473-1475 for the vanishing of the transcendence degree at closed points; load-bearing.

Source reference. Original lemma 02IZ: sources/stacks-chow/tags.txt, source lines 3318–3318. Cited at proof lines 1469-1472 to read dim(G) as the supremum of local dimensions; load-bearing.

Lemma 0.2. *Let k be a field. Let G be a locally algebraic group scheme over k. Then G is equidimensional and $\dim(G) = \dim_g(G)$ for all $g \in G$. For any closed point $g \in G$ we have $\dim(G) = \dim(\mathcal{O}_{G,g})$.*

Proof. Let us first prove that $\dim_g(G) = \dim_{g'}(G)$ for any pair of points $g, g' \in G$. By Morphisms, Lemma 02FY we may extend the ground field at will. Hence we may

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assume that both g and g' are defined over k. Hence there exists an automorphism of G mapping g to g', whence the equality. By Morphisms, Lemma 02FX we have $\dim_g(G) = \dim(\mathcal{O}_{G,g}) + \mathrm{trdeg}_k(\kappa(g))$. On the other hand, the dimension of G (or any open subset of G) is the supremum of the dimensions of the local rings of G, see Properties, Lemma 02IZ. Clearly this is maximal for closed points g in which case $\mathrm{trdeg}_k(\kappa(g)) = 0$ (by the Hilbert Nullstellensatz, see Morphisms, Section 01T9). Hence the lemma follows. $\square$